

The top pocket, tied out to the loan file

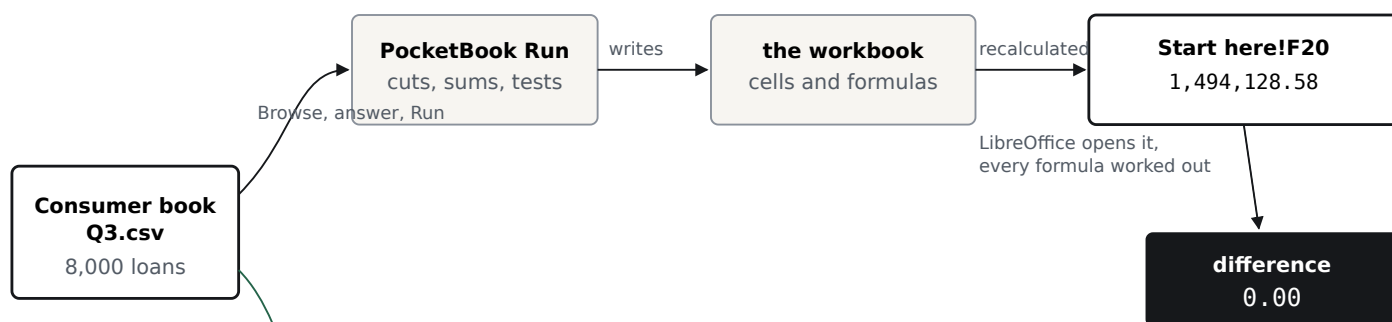
Before PocketBook goes to a bank: is the number on its front tab right? One figure, traced from the loan file to the cell the analyst reads, and worked out a second time from the same file with a plain tool that shares no code with PocketBook.

Extract: Consumer book Q3.csv (8,000 loans, 10 columns, made-up data) · Workbook: Consumer book Q3 - PocketBook.xlsx, first Run 2026-09-28 15:47 · Calculated by LibreOffice 24.2 · Hand road: awk (mawk 1.3.4), Python 3.11 csv module, scipy 1.17, statsmodels 0.15

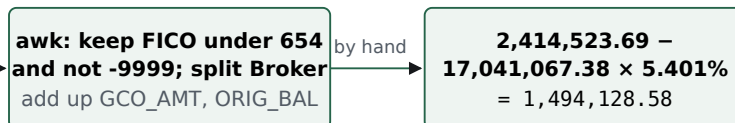
64 figures checked — 62 agree to the last digit, 1 differs, 1 could not be checked. The front-tab figure, **FICO 496 - 653 / Broker, 2.62× its band's charge-offs, \$1,494,129 above its share**, ties to the cent.

HOW THE TWO ROADS MEET

ROAD 1 · WHAT POCKETBOOK DID



ROAD 2 · THE SAME FILE, BY HAND



The same loan file travels two roads. Road 1 is PocketBook: its Run writes the workbook, and LibreOffice works out every formula when the file opens, as Excel would. Road 2 never touches PocketBook: awk filters the file's rows and adds them up, and the rest is arithmetic you can do on a calculator. **Road 2 is the one that makes this evidence:** the file cannot agree with a mistake PocketBook made while computing.

THE WORDS, SHOWN ONCE

- **Extract** — the loan file: one row per loan. Here, `Consumer book Q3.csv`.
- **Band** — a range of one number column. PocketBook cut FICO (the credit score) into five bands of about equal size. The lowest band is every real score under 654; the lowest score in it is 496, so the workbook labels it *FICO 496 - 653*.
- **Segment** — a value of a category column. Here the channel: Branch, Broker or Online.
- **Pocket** — one band crossed with one segment: *FICO 496 - 653 / Broker* is the 523 loans with a score from 496 to 653 that came through brokers.

- **Rest of band** — every other loan in the same band (the Branch and Online loans with a score under 654). The workbook compares each pocket with this, because Control says *Judged against: Rest of its band*.
- **GCO / charge-offs** — `GCO_AMT`, the gross dollars charged off on a loan. **Booked dollars** — `ORIG_BAL`, the amount lent.
- **Charge-off rate** — charge-off dollars ÷ booked dollars, for a group of loans.
- **x rest of band (the multiple)** — the pocket's rate ÷ the rest of the band's rate. 2.62x means the pocket charged off 2.62 times as much per dollar lent.
- **Dollars above share** — what the pocket charged off, less what it *would* have charged off at the rest of the band's rate: `GCO - booked × rest rate`.
- **p-value** — how likely a gap this big would be if the pocket were really no different; under 5% counts as real. **Benjamini-Hochberg** — the allowance for testing many pockets at once: each p-value is raised according to how many were tested.

FIGURE 1 • THE TOP POCKET ON START HERE

Link 1 — the figure, read out of the workbook

The analyst opens `Consumer book Q3 - PocketBook.xlsx`. Its first tab, **Start here**, lists the largest pockets that are worse and material. Row 20 is the top one.

Consumer book Q3 - PocketBook.xlsx · tab Start here · row 20 · LibreOffice Calc, recalculated on open

1	PocketBook		Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-28 15:47	
2				
3	How this tab works			
4	Where things stand		What is left before Run, counted live from Control and Columns as you fill them in.	
5	What the last Run found		Pockets worse and material at Control's lines, and the five largest.	
6	The tabs		Red tabs you fill in; black tabs hold results; grey tabs are the record.	
7				
8	WHERE THINGS STAND			
9	Answers needed before Run		Odd values to answer	
10	0		0	
11	Control and Columns		Columns · Treat as	
12			Control · Status	
13				
14				
15	WHAT THE LAST RUN FOUND			
16	Pockets worse and material, charge-offs		Dollars above their share, in those	
17	4 of 81		\$3,094,991	
18			4 short \$3,248,654	
19				
20	Largest, worse and material	Segment	Loans	× its comparison
21	FICO 496 - 653	Broker	523	2.62×
22	FICO 496 - 653	ASSET_CLASS 4	392	1.74×
23	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	1.75×
24	FICO 712 - 745	Online	564	2.06×

Start here as LibreOffice Calc shows it, the file recalculated when it opened. Ringed: row 20, and the file's own header naming the extract and the Run. Screenshot taken under a virtual display; the unmarked original is [raw/start-here.png](#).

Start here shows the pocket, its loans, the multiple and the dollars. The rates behind the multiple are on **Pockets**, which shows one measure at a time: the analyst picks *Charge-offs* in the MEASURE box (C18) to see them. The file opens on *Bad loans*, which gives the bad-loan pieces.

Consumer book Q3 - PocketBook.xlsx · tab Pockets · MEASURE (C18) set to Charge-offs · row 21									
14	LINES IN USE NOW		Worse at	Better at	How sure	Material at	Judged against	Order and "Could	
15	live from Control		1.34x	0.75x	95%	\$107,354	Rest of its band		
16									
17	MEASURE		POCKETS		SHOW				
18	Charge-offs		Two-way		All		4 worse and material · 4 worse · 41 shown		
19									
20	#	Band	Segment	Loans	This pocket	Rest of band	x rest of band	Dollars above share	Worse?
21	1	FICO 496 - 653	Broker	523	14.17%	5.40%	2.62x	1,494,129	Yes
22	2	FICO 496 - 653	ASSET_CLASS 4	392	12.32%	7.08%	1.74x	643,768	Yes
23	3	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	6.85%	3.92%	1.75x	529,198	Yes
24	4	FICO 712 - 745	Online	564	4.52%	2.19%	2.06x	427,896	Yes

Pockets with MEASURE set to Charge-offs. Row 21 is the same pocket: 523 loans, 14.17% of booked dollars charged off, the rest of its band 5.40%, 2.62×, \$1,494,129 above its share.

How the values were read. The screen shows rounded values. The full values were read with [read-workbook.py](#): it copies the workbook, has LibreOffice recalculate it through the repository's own helper ([pocketbook/tests/recalc.py](#), "recalculate on load: always"), and reads each cell with `openpyxl`. For the charge-off view it sets `Pockets!C18` to *Charge-offs* first, the one thing an analyst does. It does not import PocketBook.

```

Start here!C1      'Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-28 15:47'
Start here!B20     'FICO 496 - 653'          C20 'Broker'
Start here!D20     523                      (loans)
Start here!E20     2.6233556378895          (× its comparison)
Start here!F20     1494128.58415667          (dollars above share)
Pockets (Charge-offs) G21 0.141688524325288 H21 0.0540104140966861 I21 2.6233556378895 J21 1494128.58415667
Pockets (Bad loans)   F21 523   G21 0.244741873804971 H21 0.102713178294574 I21 2.38276994119557 J21
74.281007751938
Grids!E45          17041067.38              (booked dollars, FICO 496 - 653 / Broker)

```

Two of the pieces are not shown as numbers of their own, and are read as two cells multiplied: the pocket's bad loans are `Pockets!G21 × F21` = `0.244741873804971 × 523` = **128.0000**, and its charge-off dollars are `Pockets!G21 (Charge-offs) × Grids!E45` = `0.141688524325288 × 17,041,067.38` = **2,414,523.69**.

Link 2 — the call

The workbook came from the walk of 27 September 2026, replayed: `walk_to_first_run.py` is the walk's own driver (`docs/walkthrough/2026-09-27/driver/walk_now.py`) cut to stop after the first Run. It makes the made-up book (`synth.write_extract` , 8,000 loans), picks it with Browse, cuts FICO and ORIG_BAL into bands, segments by CHANNEL and ASSET_CLASS, splits by REV_DEBT, answers Control as the walk did (worse at the suggested line, 95% sure, materiality 1% of the book's losses, judged against the rest of its band), marks FICO –9999 as *Missing*, and presses Run. The workbook it leaves beside the extract is the one read above.

```
xvfb-run -a -s "-screen 0 1000x760x24" python3 walk_to_first_run.py ../../src OUT "/home/credit/Loan files"
```

Link 3 — the derivation, by hand

Every step is a filter or a sum a person could do in a spreadsheet:

- Which loans are in the band.** A real FICO score under 654. Leave out `-9999` (the analyst told PocketBook it means *missing*: Columns row 14) and blank scores. The band edges 654; 686; 712; 746 are printed on Record row 16.
- Which are the pocket.** Of those, `CHANNEL = Broker` . The rest of the band is the ones that are not.
- What is left out.** For charge-offs, a loan whose `GCO_AMT` is not a number or whose `ORIG_BAL` is blank (Record row 70). None of the 1,556 loans in this band is either, so nothing is left out here. For bad loans, a `BAD_FLAG` that is neither 0 nor 1: one loan in the rest of the band carries a 2, so the rest has 1,032 loans with a flag, not 1,033.
- Add up.** Loans, loans with `BAD_FLAG = 1` , `GCO_AMT` and `ORIG_BAL` , for the pocket and for the rest.
- Divide and subtract.**

	POCKET (BROKER)	REST OF BAND
Loans	523	1,033
Loans with a 0/1 flag · bad	523 · 128	1,032 · 106
Bad-loan rate	$128 \div 523 = 24.474\%$	$106 \div 1,032 = 10.271\%$
Charge-off dollars (GCO_AMT)	2,414,523.69	1,820,522.82
Booked dollars (ORIG_BAL)	17,041,067.38	33,706,885.06
Charge-off rate	14.1689%	5.4010%

- Multiple: $14.1689\% \div 5.4010\% = \mathbf{2.6234\times}$
- Its share at the rest's rate: $17,041,067.38 \times 5.40104\% = 920,395.11$
- Above its share: $2,414,523.69 - 920,395.11 = \mathbf{1,494,128.58}$
- Bad loans above share: $128 - 523 \times 10.2713\% = \mathbf{74.28}$

Link 4 — the independent source

The extract itself, `Consumer book Q3.csv`, read with `awk`: a Unix tool older than PocketBook that knows nothing about it. A copy of the file is kept beside this document, so anyone can run the same lines.

The extract, read with awk · Consumer book Q3.csv · 8,000 loans + 1 header row · 10 columns									
<code>\$ wc -l "Consumer book Q3.csv"</code>									
8001 Consumer book Q3.csv									
<code>\$ head -1 "Consumer book Q3.csv"</code>									
LOAN_NBR,FICO,CHANNEL,ORIG_BAL,BAD_FLAG,GCO_AMT,RANR_AMT,ASSET_CLASS,REV_DEBT,ORIG_DATE									
<code>\$ awk -F, 'NR>1 && \$2!="" && \$2!=-9999 && \$2<654 && \$3=="Broker"' "Consumer book Q3.csv" head -5</code>									
L0000007,649,Broker,20836.58,0,0.0,4310.88,1,15250.74,2024-03-18									
L0000011,595,Broker,20652.75,0,0.0,4713.94,4,0.0,2024-08-05									
L0000021,644,Broker,38125.36,1,27620.62,-20477.36,4,22239.77,2024-07-03									
L0000029,650,Broker,39323.65,0,0.0,13355.49,1,16145.24,2022-08-04									
L0000047,652,Broker,22518.91,1,12993.98,-7796.57,3,18859.59,2023-11-07									
<code>\$ awk -F, 'NR>1 && \$2!="" && \$2!=-9999 && \$2<654 && \$3=="Broker"' "Consumer book Q3.csv" wc -l</code>									
523									
<code>\$ awk -F, 'NR>1 && \$2!="" && \$2!=-9999 && \$2<654 {print \$2}' "Consumer book Q3.csv" sort -n sed -n '1p;\$p'</code>									
496									
653									
<code>\$ awk -F, 'NR>1 && \$2!="" && \$2!=-9999 && \$2<654 {s=(\$3=="Broker")?"Broker":"others"; n[s]++; if(\$5==0 \$5==1){t[s]++; b[s]+=\$5}; g[s]+=\$6; k[s]+=\$4} END{for(s in n) printf "%-7s loans %5d flag 0/1 %5d bad %4d GCO_AMT %13.2f ORIG_BAL %14.2f\n", s, n[s], t[s], b[s], g[s], k[s]}' "Consumer book Q3.csv" sort</code>									
Broker	loans	523	flag 0/1	523	bad 128	GCO AMT	2414523.69	ORIG BAL	17041067.38
others	loans	1033	flag 0/1	1032	bad 106	GCO AMT	1820522.82	ORIG BAL	33706885.06

Every command here was run on the file and its real output drawn beneath it. The file has 8,001 lines: one header and 8,000 loans. The filter keeps a real score under 654; the band runs from 496 to 653. The pocket is 523 rows; ringed, its bad loans, charge-off dollars and booked dollars, and the rest of the band's.

The extract, read with awk · Consumer book Q3.csv · Figure 1 worked through

```
$ ./by-hand.awk "Consumer book Q3.csv" # the Figure 1 part of its output
rows read: 8000
```

```
FIGURE 1 -- FICO under 654 (lowest score seen 496, highest 653), Broker vs the rest of the band
pocket loans 523 | flag 0/1 523, bad 128, bad rate 0.2447418738
charge-off loans 523, GC0 2414523.69, booked 17041067.38, GC0/booked 0.1416885243
rest loans 1033 | flag 0/1 1032, bad 106, bad rate 0.1027131783
charge-off loans 1033, GC0 1820522.82, booked 33706885.06, GC0/booked 0.0540104141
charge-offs: multiple = 0.1416885243 / 0.0540104141 = 2.6233556379
its share at the rest's rate = 17041067.38 x 0.0540104141 = 920395.1058
above its share = 2414523.69 - 920395.1058 = 1494128.5842
bad loans: multiple = 0.2447418738 / 0.1027131783 = 2.3827699412
bad loans above share = 128 - 523 x 0.1027131783 = 74.281008
```

The same sums carried through to the rate, the multiple and the dollars (by-hand.awk, the Figure 1 part of its output).
 Ringed: the rest of the band's rate, the multiple, the dollars above share, and the bad loans above share.

Link 5 — the comparison

	workbook (cell, as calculated)	extract (by hand)	diff
loans	Start here!D20 523	523	0
bad loans	Pockets!G21xF21 128.0000	128	0.0000
bad-loan rate, pocket	Pockets!G21 0.244741873805	0.244741873805	0
bad-loan rate, rest of band	Pockets!H21 0.102713178295	0.102713178295	0
bad-loan multiple	Pockets!I21 2.382769941196	2.382769941196	0
bad loans above share	Pockets!J21 74.281007751938	74.281008	0.000000
booked dollars	Grids!E45 17,041,067.38	17,041,067.38	0.00
charge-off dollars	G21xGrids!E45 2,414,523.69	2,414,523.69	0.00
charge-off rate, pocket	Pockets!G21 0.141688524325	0.141688524325	0
charge-off rate, rest of band	Pockets!H21 0.054010414097	0.054010414097	0
x its comparison	Start here!E20 2.623355637890	2.623355637890	0
dollars above share	Start here!F20 1,494,128.58	1,494,128.58	0.00

Same loans (the same file, 8,000 rows), same run (15:47, 28 Sep 2026), same basis (gross charge-offs over the amount lent, against the rest of the band), same units (dollars, no scaling). **TIED** on all twelve.

FIGURE 2 · "702: EVERY GRID ADDS UP TO THE BOOK"

Link 1 — the figure

Record says, under *Does it add up*: Tie-out checks: 702: every grid adds up to the book (Record!C25). One grid is picked to prove the claim rather than the count: the one **Grids** opens on, *FICO x CHANNEL*. Its Loans block puts every loan in one cell; its booked-dollars table does the same for the amount lent.

Consumer book Q3 - PocketBook.xlsx · tab Record · Does it add up, C25		
8	This Run	
9	Item	Detail
10	Extract	Consumer book Q3.csv
11	Loans run	8,000
12	What was run	Where the book bleeds
13	Record of this run	Consumer book Q3 - PocketBook - what ran.yaml, beside this workbook: every setting the run used, kept for the
14		file. It is replaced by the next Run.
15	Origination dates	2021-06-30 to 2026-03-30 (8,000 loans; 0 without a readable date)
16	Band edges used: FICO	654; 686; 712; 746 (5 bands)
17	Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
18	Split	REV_DEBT, each pocket halved at its own median. REV_DEBT isn't cut on its own while it splits. Split pockets:
19		158.
20	How closely REV_DEBT moves with FICO	correlation -0.53
21	How closely REV_DEBT moves with ORIG_BAL	correlation -0.02
22		
23	Does it add up	
24	Check	Result
25	Tie-out checks	702: every grid adds up to the book
26	Worse now: Bad loans	7 of 81 pockets on the grids read worse; 7 of them are material.
27	Worse now: Bad dollars	6 of 81 pockets on the grids read worse; 6 of them are material.
28	Worse now: Charge-offs	4 of 81 pockets on the grids read worse; 4 of them are material.

Record, ringed at C25.

Consumer book Q3 - PocketBook.xlsx · tab Grids · GRID FICO x CHANNEL · the Loans block, M27:Q35

12	GRID	MEASURE			
13	FICO x CHANNEL	Bad loans			

Loans				
FICO	Branch	Broker	Online	All
496 - 653	502	523	531	1,556
654 - 685	503	531	505	1,539
686 - 711	515	501	546	1,562
712 - 745	538	509	564	1,611
746 - 921	518	536	517	1,571
(blank)		1		1
(marked missing)	51	54	55	160
All	2,627	2,655	2,718	8,000

Grids on FICO x CHANNEL: the GRID box (rows 12-13) above, and, from the right-hand side of the same screen, the Loans block M27:Q35, ringed — seven bands down, three channels across, totals in All. The full screen is [raw/grids-loans.png](#).

12	GRID				MEASURE			
13	FICO x CHANNEL				Bad loans			
38								
39	How common each group is: a count, not a test							
40	REV_DEBT, each pocket cut at its own median							
41								
42	FICO	CHANNEL	Loans	Booked dollars	High half	Low half		
43					Loans	Booked dollars	Loans	Booked dollars
44	496 - 653	Branch	502	16,339,707	251	8,045,394	251	8,294,313
45	496 - 653	Broker	523	17,041,067	261	8,288,108	262	8,752,960
46	496 - 653	Online	531	17,367,178	265	8,466,411	266	8,900,767
47	654 - 685	Branch	503	16,176,091	251	7,704,192	252	8,471,899
48	654 - 685	Broker	531	17,321,902	265	8,652,360	266	8,669,543
49	654 - 685	Online	505	15,851,603	252	7,627,316	253	8,224,286
50	686 - 711	Branch	515	16,839,795	257	8,275,965	258	8,563,831
51	686 - 711	Broker	501	16,516,468	250	8,314,330	251	8,202,137
52	686 - 711	Online	546	17,660,165	273	9,012,046	273	8,648,118
53	712 - 745	Branch	538	16,513,064	269	8,471,219	269	8,041,845
54	712 - 745	Broker	509	16,230,509	254	8,019,072	255	8,211,437
55	712 - 745	Online	564	18,368,910	282	9,087,810	282	9,281,101
56	746 - 921	Branch	518	16,450,330	259	8,314,871	259	8,135,459
57	746 - 921	Broker	536	17,135,666	268	8,093,973	268	9,041,694
58	746 - 921	Online	517	16,939,748	258	8,792,490	259	8,147,258
59	(blank)	Broker	1	27,999	0	0	1	27,999
60	(marked missing)	Branch	51	1,771,004	25	798,277	26	972,727
61	(marked missing)	Broker	54	1,668,510	27	835,771	27	832,739
62	(marked missing)	Online	55	1,848,729	27	858,062	28	990,667
63	Every pocket		8,000	258,068,446	3,994	127,657,667	4,006	130,410,779
64	Share of the grid				49.9%	49.5%	50.1%	50.5%

Further down the same tab: loans and booked dollars for each of the 19 pockets, D44:E63, adding to 8,000 loans and \$258,068,446.

Links 2 and 3 — the call and the derivation

The same Run as Figure 1. By hand: put every row of the extract in exactly one cell — its FICO band (or *blank*, or *marked missing* for –9999) by its channel — and count; add ORIG_BAL the same way, a blank as 0. Every row lands in one cell, so the cells add to the file by construction; the check is whether the workbook's cells are the same cells.

Link 4 — the independent source

The extract, read with awk · Consumer book Q3.csv · the FICO x CHANNEL grid				
\$./by-hand.awk "Consumer book Q3.csv" # the Figure 2 part of its output				
FIGURE 2 -- the FICO x CHANNEL grid: loans in each cell (booked dollars beneath)				
FICO band	Branch	Broker	Online	All
1 <654	502	523	531	1556
	16339706.79	17041067.38	17367178.27	
2 654-685	503	531	505	1539
	16176090.76	17321902.42	15851602.50	
3 686-711	515	501	546	1562
	16839795.27	16516467.51	17660164.64	
4 712-745	538	509	564	1611
	16513064.03	16230509.40	18368910.49	
5 746+	518	536	517	1571
	16450329.83	17135666.28	16939748.20	
(blank)	0	1	0	1
	0.00	27999.47	0.00	
(marked missing)	51	54	55	160
	1771003.54	1668510.01	1848729.24	
All	2627	2655	2718	8000
booked dollars, every loan:	258068446.03			

by-hand.awk on the extract, the Figure 2 part of its output: loans in each cell, booked dollars beneath. Ringed: the lowest band's total, the grand total of 8,000 loans, the pocket from Figure 1, and \$258,068,446.03 booked across every loan.

Link 5 — the comparison

Loans, workbook (M27:Q35) against extract, cell by cell. Each cell shows workbook / extract ; every difference is 0.

FICO BAND	BRANCH	BROKER	ONLINE	ALL
496 - 653	502 / 502	523 / 523	531 / 531	1,556 / 1,556
654 - 685	503 / 503	531 / 531	505 / 505	1,539 / 1,539
686 - 711	515 / 515	501 / 501	546 / 546	1,562 / 1,562
712 - 745	538 / 538	509 / 509	564 / 564	1,611 / 1,611
746 - 921	518 / 518	536 / 536	517 / 517	1,571 / 1,571
(blank)	— / 0	1 / 1	— / 0	1 / 1
(marked missing)	51 / 51	54 / 54	55 / 55	160 / 160
All	2,627 / 2,627	2,655 / 2,655	2,718 / 2,718	8,000 / 8,000

Booked dollars, workbook (E44:E63) against extract, to the cent:

POCKET	WORKBOOK	EXTRACT	POCKET	WORKBOOK	EXTRACT
496-653 Branch	16,339,706.79	16,339,706.79	712-745 Branch	16,513,064.03	16,513,064.03
496-653 Broker	17,041,067.38	17,041,067.38	712-745 Broker	16,230,509.40	16,230,509.40
496-653 Online	17,367,178.27	17,367,178.27	712-745 Online	18,368,910.49	18,368,910.49
654-685 Branch	16,176,090.76	16,176,090.76	746-921 Branch	16,450,329.83	16,450,329.83
654-685 Broker	17,321,902.42	17,321,902.42	746-921 Broker	17,135,666.28	17,135,666.28
654-685 Online	15,851,602.50	15,851,602.50	746-921 Online	16,939,748.20	16,939,748.20
686-711 Branch	16,839,795.27	16,839,795.27	(blank) Broker	27,999.47	27,999.47
686-711 Broker	16,516,467.51	16,516,467.51	missing Branch	1,771,003.54	1,771,003.54
686-711 Online	17,660,164.64	17,660,164.64	missing Broker	1,668,510.01	1,668,510.01
missing Online	1,848,729.24	1,848,729.24	Every pocket	258,068,446.03	258,068,446.03

TIED on all 30 loan cells and all 20 dollar cells. The workbook leaves an empty cell blank where the extract has 0 (the *blank* FICO row has no Branch or Online loan): same count, shown two ways.

The count itself, **702**, is **COULD NOT**: it is the number of checks PocketBook ran on itself, and the extract has nothing to say about how many checks there should be. What the count claims — every grid adds up to the book — was tested above on one grid of the eight.

FIGURE 3 · THE POCKET'S P-VALUE

Link 1 — the figure

On Pockets, with the file as it opens (MEASURE *Bad loans*), row 21 carries the pocket's p-value in L21. The screen says **under 0.01%**; the cell holds **2.4384791384469e-12** (0.0000000000024).

Consumer book Q3 - PocketBook.xlsx · tab Pockets · MEASURE Bad loans (as it opens) · row 21, p-value in L21; row 37

14

LINES IN USE NOW

Worse at

Better at

How sure

Material at

judged against

15

live from Control

1.34x

0.75x

95%

6.2 bad loans

Rest of its band

16

17

MEASURE

POCKETS

SHOW

18

Bad loans

Two-way

All

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

Order and "Could have caught" are from the last Run.

#	Band	Segment	Loans	This pocket	Rest of band	x rest of band	Bad loans above share	Worse?	p-value	Material?
1	FICO 496 - 653	Broker	523	24.47%	10.27%	2.38x	74.3	Yes	under 0.01%	Yes
2	FICO 712 - 745	Online	564	8.69%	3.63%	2.39x	28.5	Yes	0.02%	Yes
3	FICO 496 - 653	ASSET_CLASS 4	392	20.41%	13.24%	1.54x	28.1	Yes	1.4%	Yes
4	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	12.32%	7.00%	1.76x	22.0	Yes	1.5%	Yes
5	FICO 654 - 685	ASSET_CLASS 4	410	11.71%	7.26%	1.61x	18.2	Yes	4.5%	Yes
6	ORIG_BAL 26,324 - 37,950	ASSET_CLASS 4	412	9.95%	5.82%	1.71x	17.0	Yes	4.3%	Yes
7	FICO 712 - 745	ASSET_CLASS 4	427	8.20%	4.39%	1.87x	16.2	Yes	3.4%	Yes
8	ORIG_BAL 15,449 - 26,323	Broker	537	11.73%	8.09%	1.45x	19.6	Not sure	11.0%	Yes
9	ORIG_BAL 49,152 - 59,984	Broker	520	8.46%	5.46%	1.55x	15.6	Not sure	11.0%	Yes
10	FICO 686 - 711	Online	546	7.69%	5.31%	1.45x	13.0	Not sure	16.0%	Yes
11	ORIG_BAL 15,449 - 26,323	ASSET_CLASS 4	413	11.38%	8.59%	1.32x	11.5	No	23.3%	Yes
12	ORIG_BAL 5,005 - 15,448	ASSET_CLASS 4	425	9.65%	6.98%	1.38x	11.3	Not sure	21.9%	Yes
13	ORIG_BAL 49,152 - 59,984	ASSET_CLASS 4	384	8.59%	5.76%	1.49x	10.9	Not sure	19.3%	Yes
14	FICO 686 - 711	ASSET_CLASS 4	407	8.11%	5.45%	1.49x	10.8	Not sure	16.6%	Yes
15	ORIG_BAL 37,951 - 49,151	Broker	531	9.23%	7.95%	1.16x	6.8	No	86.7%	Yes
16	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 3	382	9.69%	7.96%	1.22x	6.6	No	44.5%	Yes
17	FICO (marked missing)	Broker	54	12.96%	2.83%	4.58x	5.5	Too few losses	9.6%	No

Pockets on Bad loans. Ringed: row 21 and its p-value, and row 37 — FICO (marked missing) / Broker, which reads *Too few losses* and still shows a p-value of 9.6%. That second ring is what the tie-out found.

Links 2 and 3 — the call and the derivation

The same Run. The method, as the workbook states it in words (Pockets!D8 and Record!F25): for bad loans, a pocket of 65 loans or more gets the **pooled two-proportion z test**, two-sided, against the rest of its band; one under 65 gets Fisher's exact test. The p-values are then adjusted by **Benjamini-Hochberg across one grid and one measure** — here FICO x CHANNEL on bad loans. Record!C47

says of this grid: "15 have at least 10 bad loans and can be tested", and Pockets!D7 says a pocket under 10 is "Too few losses ... so not tested".

By hand: 128 bad of 523 against 106 bad of 1,032. Pooled rate = $234 \div 1,555 = 15.048\%$. Standard error = $\sqrt{(0.15048 \times 0.84952 \times (1/523 + 1/1,032))} = 0.019191$. $z = (0.244742 - 0.102713) \div 0.019191 = 7.4007$. Two-sided $p = 1.3547 \times 10^{-13}$. This is the smallest p in the grid, so Benjamini-Hochberg multiplies it by the number of tests in the family.

Link 4 — the independent source

`pvalue.py` reads the extract with Python's `csv` module and uses `scipy` for the normal tail, `scipy's fisher_exact` for small pockets and `statsmodels' multipletests(method="fdr_bh")` for the adjustment. It imports nothing from PocketBook.

The extract, read with Python's `csv` module, `scipy` and `statsmodels` · Consumer book Q3.csv

```
$ python3 pvalue.py "Consumer book Q3.csv"
```

A. The pockets with at least 10 bad loans (what Record!C47 says was tested): 15 pockets in the Benjamini-Hochberg family

band	channel	bad	loans	r.bad	r.loans	test	z	p raw	p after BH
band 1	Branch	51	501	183	1054	z	-3.7020	2.138946e-04	8.021048e-04
band 1	Broker	128	523	106	1032	z	7.4007	1.354711e-13	2.032066e-12
band 1	Online	55	531	179	1024	z	-3.7252	1.951916e-04	8.021048e-04
band 2	Branch	45	503	85	1036	z	0.4908	6.235893e-01	8.063281e-01
band 2	Broker	43	531	87	1008	z	-0.3575	7.207543e-01	8.063281e-01
band 2	Online	42	505	88	1034	z	-0.1284	8.978554e-01	8.978554e-01
band 3	Branch	28	515	68	1047	z	-0.8184	4.131559e-01	6.885931e-01
band 3	Broker	26	501	70	1061	z	-1.0814	2.795095e-01	5.240803e-01
band 3	Online	42	546	54	1016	z	1.8654	6.212433e-02	1.553108e-01
band 4	Branch	22	538	65	1073	z	-1.6487	9.921359e-02	2.126005e-01
band 4	Broker	16	509	71	1102	z	-2.7238	6.452726e-03	1.935818e-02
band 4	Online	49	564	38	1047	z	4.2848	1.828876e-05	1.371657e-04
band 5	Branch	23	518	39	1053	z	0.7048	4.809302e-01	7.213952e-01
band 5	Broker	20	536	42	1035	z	-0.3152	7.525729e-01	8.063281e-01
band 5	Online	19	517	43	1054	z	-0.3871	6.987036e-01	8.063281e-01

B. Every pocket with a rest of its band to compare with: 18 pockets in the Benjamini-Hochberg family

band	channel	bad	loans	r.bad	r.loans	test	z	p raw	p after BH
(marked missing)	Branch	1	51	9	109	exact	nan	1.709099e-01	3.418197e-01
(marked missing)	Broker	7	54	3	106	exact	nan	3.183642e-02	9.550925e-02
(marked missing)	Online	2	55	8	105	exact	nan	4.960520e-01	6.868412e-01
band 1	Branch	51	501	183	1054	z	-3.7020	2.138946e-04	9.625258e-04
band 1	Broker	128	523	106	1032	z	7.4007	1.354711e-13	2.438479e-12
band 1	Online	55	531	179	1024	z	-3.7252	1.951916e-04	9.625258e-04
band 2	Branch	45	503	85	1036	z	0.4908	6.235893e-01	7.968419e-01
band 2	Broker	43	531	87	1008	z	-0.3575	7.207543e-01	7.968419e-01
band 2	Online	42	505	88	1034	z	-0.1284	8.978554e-01	8.978554e-01
band 3	Branch	28	515	68	1047	z	-0.8184	4.131559e-01	6.760732e-01
band 3	Broker	26	501	70	1061	z	-1.0814	2.795095e-01	5.031171e-01
band 3	Online	42	546	54	1016	z	1.8654	6.212433e-02	1.597483e-01
band 4	Branch	22	538	65	1073	z	-1.6487	9.921359e-02	2.232306e-01
band 4	Broker	16	509	71	1102	z	-2.7238	6.452726e-03	2.322982e-02
band 4	Online	49	564	38	1047	z	4.2848	1.828876e-05	1.645989e-04
band 5	Branch	23	518	39	1053	z	0.7048	4.809302e-01	6.868412e-01
band 5	Broker	20	536	42	1035	z	-0.3152	7.525729e-01	7.968419e-01
band 5	Online	19	517	43	1054	z	-0.3871	6.987036e-01	7.968419e-01

Run on the extract. Family A is the 15 pockets the workbook says were tested: the pocket's adjusted p -value is 2.032066e-12 (ringed). Family B adds the three *marked missing* pockets that have too few losses: 2.438479e-12 (ringed), which is the workbook's figure — and the marked-missing Broker pocket's 9.550925e-02 (ringed) is the 9.6% on row 37.

Link 5 — the comparison

Pockets!L21, as calculated	2.4384791384469e-12	
by hand, the method the workbook describes (15 diff large)	2.032066e-12	+0.406413e-12 (the workbook's is 18/15 = 1.2 times as large)
by hand, counting all 18 pockets with a p-value diff	2.438479e-12	0.000000e-12

DIFFERS against the method the workbook describes. The raw test is right: 1.354711e-13 on both roads. The gap is entirely in **how many pockets the allowance counts** — 18 in the workbook's arithmetic, 15 in its words. See *What it found*, item 2.

THE ROSTER

VERDICT	FIGURES WHICH, AND AGAINST WHAT
DIFFERS	1 Figure 3: the pocket's p-value, Pockets!L21, 2.44e-12 against 2.03e-12 by the method the workbook states. Denominator: the 15 pockets Record!C47 calls testable. It ties exactly with 18.
COULD NOT	1 Record!C25's count of 702 tie-out checks. Obstacle: it counts checks PocketBook ran on itself; no file outside PocketBook says how many there should be. Its claim was tested on one grid instead (below).
TIED	12 Figure 1, the top pocket: loans, bad loans, both bad-loan rates, the bad-loan multiple, bad loans above share, booked dollars, charge-off dollars, both charge-off rates, the 2.62× multiple and the \$1,494,128.58 above share. Denominator: its 523 loans against the band's other 1,033.
TIED	30 Figure 2, the FICO x CHANNEL Loans block: 19 pocket cells, 7 band totals, 3 channel totals and the 8,000 in all. Denominator: the extract's 8,000 rows.
TIED	20 Figure 2, booked dollars: the 19 pockets and the \$258,068,446.03 in all, to the cent.

Checked: 64. Tied: 62 of 64. One grid of the workbook's eight, one pocket of its 239, one p-value of its 2,684 tests.

HOW TO RUN IT YOURSELF

Everything here runs from the folder this document sits in: `pocketbook/docs/tie-out/2026-09-28/`. The extract and the workbook are both in it.

Step 1 • Count the file and the pocket's rows. Expect `8001` and `523`.

```
wc -l "Consumer book Q3.csv"
awk -F, 'NR>1 && $2!=" " && $2!=-9999 && $2<654 && $3=="Broker"' "Consumer book Q3.csv" | wc -l
```

Step 2 • Add up the band, Broker against the rest. Expect Broker 523 loans, 128 bad, 2414523.69 charged off, 17041067.38 booked; others 1,033 loans (1,032 with a 0/1 flag), 106 bad, 1820522.82 and 33706885.06.

```
awk -F, 'NR>1 && $2!=" " && $2!=-9999 && $2<654 {s=($3=="Broker")?"Broker":"others"; n[s]++; if($5==0||$5==1) {t[s]++; b[s]+=$5}; g[s]+=$6; k[s]+=$4} END{for(s in n) printf "%-7s loans %5d flag 0/1 %5d bad %4d GCO_AMT %13.2f ORIG_BAL %14.2f\n", s, n[s], t[s], b[s], g[s], k[s]}' "Consumer book Q3.csv"
```

Step 3 • The dollars above share, on a calculator. $2,414,523.69 - 17,041,067.38 \times (1,820,522.82 \div 33,706,885.06) = \mathbf{1,494,128.58}$. The multiple: $(2,414,523.69 \div 17,041,067.38) \div (1,820,522.82 \div 33,706,885.06) = \mathbf{2.6234}$.

Step 4 • Figures 1 and 2 in one go.

```
awk -f by-hand.awk "Consumer book Q3.csv"
```

Step 5 • The p-value (needs scipy and statsmodels). Family A gives 2.032066e-12, family B 2.438479e-12.

```
python3 pvalue.py "Consumer book Q3.csv"
```

Step 6 • The workbook's side. Open `Consumer book Q3 - PocketBook.xlsx` in Excel and read Start here row 20; on Pockets pick *Charge-offs* in C18 and read row 21. Or, from a copy of the repository with LibreOffice installed:

```
python3 read-workbook.py "Consumer book Q3 - PocketBook.xlsx"
```

Step 7 • Make the book, the workbook and this document again from nothing (optional; the screenshots need a virtual display).

```
xvfb-run -a -s "-screen 0 1000x760x24" python3 walk_to_first_run.py ../../src OUT "/home/credit/Loan files"
xvfb-run -a -s "-screen 0 1600x1300x24" python3 workbook-shots.py
xvfb-run -a -s "-screen 0 2200x1300x24" python3 workbook-shots.py pockets-badloans grids-loans
python3 pictures.py
python3 build.py
```

WHAT IT FOUND

- 1. The headline is right.** *FICO 496 - 653 / Broker, 2.62× its band's charge-offs, \$1,494,129 above its share* ties to the cent, and so does every piece under it. The walk's procedure (27 Sep) printed the same numbers; this run reproduced them from a fresh book. (The brief for this job quoted the band as 596 to 653; the walk's document and the workbook both say **496**.)
- 2. The p-value allowance counts pockets the workbook says it did not test.** Pockets!D7 says a pocket with fewer than 10 bad loans reads *Too few losses ... so not tested*, and Record!C47 says 15 pockets of this grid can be tested. But those pockets are still given a p-value (row 37: 9.6%) and still counted in Benjamini-Hochberg's family, which therefore holds 18, not 15. Every bad-loan p-value on this grid comes out up to 1.2 times what the stated method gives. Where it is in the code:
 - `pocketbook/src/pocketbook/engine.py:1148` works out `p_band` for every pocket with a rest of its band, with no floor on bad loans;
 - `pocketbook/src/pocketbook/engine.py:1171-1173` adjusts over every cell that has a p-value, so the too-few pockets join the family.

Fixed the same day, and re-checked. The firm had already said a pocket with too few losses is not tested, so the code was brought to the tab's words: such a pocket now has no p-value and is not counted in the allowance (`engine.py`, `_judge`; test

`test_a_pocket_with_too_few_losses_has_no_p_value_and_is_not_counted_in_the_allowance`, with a

planted bug it catches). The walk was replayed on the fixed build with the same extract (identical but for line endings) and the workbook read the same way: **Pockets!L21 =**

2.03206594870575e-12, against **2.032066e-12** from the extract by `pvalue.py`'s first reading (15 pockets). That figure now ties. The roster below records the build this document was first run on, where it differed.

It errs on the careful side — p-values higher, not lower — and on this grid it changes no verdict at 5% (the one nearest the line, FICO 712 – 745 / Broker, goes from 1.94% to 2.32%, and it is a *better* pocket). It was not checked on the other seven grids or the dollar measures. For this pocket the screen reads *under 0.01%* either way, so an analyst would never see it. **Not fixed here:** whether a pocket too small to test should count against the others is the firm's call, and then either the words or the family should change so they agree.

3. **Two figures a banker will ask for are not on the page.** The pocket's bad-loan count (128) and its charge-off dollars (\$2,414,524) appear nowhere as numbers; they have to be worked out from a rate times loans or booked dollars. Both tie. A reader walking a banker through the pocket will want them in a cell.

WHAT I GOT WRONG

- **I nearly reported the p-value as a PocketBook arithmetic error.** My first run used the 15 pockets the workbook says it tested and landed on 2.03e-12. The ratio to the workbook's figure was exactly $1.2 = 18 \div 15$, which pointed at the size of the family rather than the test. Adding the three untested pockets tied it to every digit, and the marked-missing pocket's 9.6% tied too. The number is computed correctly; what it is computed *over* disagrees with the tab's own words.
- **My first workbook carried a path nobody should see.** The first replay kept the loan file in a working folder of this session, and the workbook records where its extract came from, so that folder's name was written inside the file. The walk was run again with the loan file in `/home/credit/Loan files`; the extract came out byte for byte the same, and every figure read out of the new workbook matched the first one except the Run's time (15:28 then, 15:47 now). The workbook beside this document is the second.
- **My first workbook screenshots missed the columns that mattered.** The Pockets p-value (column L) and the Grids Loans block (columns M to Q) were off the right edge of a 1,600-pixel screen, and an early shot of Start here scrolled the frozen top row out of view, hiding the line that names the extract. They were retaken wider and from the top; the originals in `raw/` are the retakes.

WHAT THIS DOES NOT PROVE

- **Not Excel.** The workbook's formulas were worked out by LibreOffice 24.2, which is what the repository's tests use. Excel was not run. A formula that LibreOffice and Excel read differently would not show here.
- **Not a real extract.** The loan file is made-up data from PocketBook's own generator (`synth.py`, 8,000 loans, a pocket planted in it). That does not make it a mirror — the question is whether the workbook reports what is in the file, and a file cannot agree with a mistake in the arithmetic — but a real bank extract has odd values this one does not: quoted fields with commas in them, dates in

other shapes, codes other than –9999. The awk lines split on every comma and would be wrong on a quoted field.

- **Not the bank's machine.** Everything ran on Linux. The bank machine is Windows with Excel; the checklist for it is `docs/BANK-MACHINE-CHECKLIST.pdf`, and none of it was exercised here.
- **Not the rest of the workbook.** One pocket of 239, one grid of eight, one p-value. The dollar measures' p-values come from shuffling loans 10,000 times at random and were not recomputed. The split, scouting and the new-variables joint model (its odds ratios) were not tied out.
- **Not the 702.** The count of checks PocketBook ran on itself was taken on trust; one grid's worth of what those checks claim was tested.